

Anne Arundel Community College
Department of Mathematics
MAT 202-Linear Algebra Course Syllabus
Section 001

Course information Term: Spring 2024 Class meets on Tues and Thurs, 8-9:45am, in MATH 204 Course credit hours: 4.0 credits	Instructor Information Name: Thao-Nhi Luu E-mail: Canvas email (preferred) or tluu@aacc.edu
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Communication: The best way to contact me is through Canvas email. I will respond to all student emails within 36 hours (Monday – Saturday)

Course Description: Study vector spaces and linear transformations on finite-dimensional vector spaces. Topics include linear systems, matrices, determinants, inner product spaces, and eigenvalues. Students may not earn credit for MAT202 and MAT202H.

Prerequisites: MAT 191 or equivalent

Course Purpose within a Program of Study: This is one of the courses in the program requirement for the A.S. Mathematics and Computer Science degrees. Check with your advisor to see if this course is counted toward your degree program.

Course Attendance/Participation Policy

- Please arrive to class on-time. *Attendance is extremely important.* Missing even one class can significantly impact your learning. Class time will be devoted to homework problems, lecture, and small group discussions. Please ask questions when you do not understand something. Students are responsible for material covered on the days they miss class.
- Regular attendance and active participation promote student success. You are also invited to participate in our Tutoring/Help Sessions.

Academic Attendance Procedure: Faculty are required to report attendance for financial aid purposes. Please refer to the link below to see detailed information regarding the Academic Attendance Reporting Procedure. Students should be aware that lack of attendance can affect financial aid.

<https://www.aacc.edu/policies/class-attendance-reporting/>

Course Outline: The course outline provides the due dates for all graded learning activities. (See separate document for the outline). For a typical section in the outline, you will...

1. read the section from the course text and read any additional provided notes prior to attending class
2. print out the Guided Learning notes (GLN) to take notes in class and participate in classroom discussion.
3. complete the corresponding Homework Assignment in WebAssign.

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As you work through those items, check with the upcoming Written Assignment problems to see if any of those questions are related to the section. Don't wait until the due dates to work on these assignments. Start on them as soon as possible. Write up your solutions to a Written Assignment *at least one* day before its due date.

Reading: You are strongly encouraged to read our course textbook as we work through the chapters. There are many reasons why reading our course text can help you to succeed in course. Reading the material we cover in class from a different source can be an effective method to reinforce your understanding.

Tutoring/Help Sessions: These hours are your time to meet with me (and your classmates) to ask any questions, receive & share feedback, learn about opportunities, and develop our class community. Please do not hesitate to stop by to discuss your questions or just to say hi.

- **Tuesdays:**
 - 11:45am-12:20pm in CRSC 190. This is an in-person only session.
 - 12:30-1:30pm in LIB 104. You are welcomed to join me in Zoom during this time, but you will need to let me know by 12:30pm that day.
- **Wednesday:**
 - 6:00-7:50pm in Zoom. This is a virtual only session.
- **Thursdays:**
 - 11:45am-12:20pm in CRSC 190. This is an in-person only session.
 - 12:30-1:30pm in LIB 104. You are welcomed to join me in Zoom during this time, but you will need to let me know by 12:30pm that day.
- **Fridays:**
 - 8:00-9:50am in Zoom. This is a virtual only session

You are invited to reach out for an appointment if the times listed above do no work with your schedule. I often need 48 hours to rearrange my schedule, but please feel free to reach out the same day. If I can meet, I will let you know right away. The Zoom link below will be used for all virtual tutoring/help sessions and for any synchronous meeting appointment:

<https://aacc-edu.zoom.us/j/94709290474?pwd=MXIKS09sMHRzNEpnZk1wNFJndGhXQT09>

Meeting ID: 947 0929 0474

Passcode: Hypatia

Resources and support: Please do not wait until the week we have an exam to start getting help. I am available throughout the week to answer questions about the material presented in the videos or the homework assignments. Please see the Getting Started Module of our Canvas course for a full list of available resources and support.

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Learner Support Services (LSS): Any student who faces challenges securing their food, transportation, housing, or course materials and believes this may affect their performance in this course, is urged to contact the Office of the Dean of Student Engagement for support. Please email deanstudents@aacc.edu or call 410-777-2582.

Required Text and Equipment

1. Textbook: Elementary Linear Algebra, 8th Edition, by Ron Larson, 2017 with access to WebAssign. Publisher: Cengage Learning. The standalone access code to WebAssign gives you access to the e-book, your homework assignments, and other resources. The physical textbook is not required, but you can purchase a bookstore bundle which includes a loose-leaf copy of the text and a WebAssign access code.
2. Calculators: Students are allowed to use a handheld calculator for testing. A TI-84 or TI-83+ are recommended. Calculators that use a computer algebra system are not permitted. Calculators may not be shared during exams. Cell phones may not be used as calculators in class
3. This is an **in-person** course. However, you are also welcomed to join me during my synchronous Tutoring/Help Sessions. You will need to have access to a computer with a reliable internet connection, a working webcam, and a microphone when we meet virtually. I also encourage you to use earphones or headphones if external noise (siblings, parents, dogs, cats, etc.) provides a challenge. For more information about resources for online learning, visit the Virtual Campus Orientation site via the link found on the main navigation menu in the course
4. You will need software and/or hardware to either type or write and scan your work and submit it electronically in Canvas. There are many free scanning apps available on Android and iOS platforms. Important requirements regarding scans:
 - All scans must be uploaded in .pdf format without exception.
 - Scans of multiple pages must be combined into a single .pdf document. Most free scanning apps have the capability to create a single document from multiple scans. If you need help with this, just let me know and we can figure it out together.
5. If you do not have a webcam or a laptop:
 - The AACC library has a limited number of standalone webcams and laptops to borrow for FREE. To request these items, email the library at circdesk@aacc.edu from your myAACC email. In the email, provide your full name, courses you are taking, the semester, as well as the item(s) you are requesting to rent (laptop or webcam).
 - If you are receiving Financial Aid, funds can be used to purchase a webcam from the AACC Bookstore.

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Graded Assignments and Learning Activities

Two midterm exams and the final exam: These exams are mandatory and proctored in class. The following are two grade distribution options for the midterm exams and the final exam, which ever better is better for you at the end of the semester.

- Option 1: Each midterm exam determines 25% of the course grade and the final exam determines 25% of the course grade.
- Option 2: One midterm exam determines 10%, the other midterm exam determines 25%, and the final determine 40% of the course grade.

If you have a conflict with a scheduled exam, reach out to me right away and at least 10 days prior to the exam to reschedule the exam.

If you missed a scheduled exam due to an unforeseen emergency, you would need to make it up with in 48 hours of the original due date and time. A documented emergency is required if you need to make up the exam after the 48-hour grace period.

Written Assignments: The average of these written assignments will determine 12% of your course grade. These written assignments are due on Saturdays. You have an automatic 24-hour grace period for each written assignment. This means you can submit your written assignments on Sundays without penalty. Beyond this grace period, you will incur a 2-point deduction for each day your assignment is late.

Writing out arguments and detailed computations is an important part of learning and understanding mathematics. These assignments give you a chance to write out and share complete solutions for a smaller set of problems, and give me a chance to provide you with some feedback for your learning. Furthermore, these written assignments will be helpful practice for you to prepare for exams.

You are invited to discuss these questions with me and/or your classmates. Our scheduled tutoring/help sessions and our Canvas Discussion boards are perfect places for these discussions. You are also welcomed to reach out for an appointment as needed.

Homework Assignments: These assignments are administered in WebAssign. Your average of these assignments determines 13% of your course grade. I will drop the two lowest homework assignments at the end of the course.

The due date for each homework assignment is listed in the course outline and in WebAssign. If you miss a due date for a homework assignment, you have 24 hours from the due time to request and extension. Once you request an extension, you have 24 hours from time of request to submit the assignment. For example, if an assignment is due at 11:59pm on Tuesday, you have until 11:59pm on Wednesday to request an extension. The Extension window closes 2 days after the original due time.

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WebAssign homework assignments give you an important opportunity to practice the skills you learn from our class, and help you assess your understanding. However, you can and should always strengthen your skills by doing additional problems from our course text. At the end of each section of the text, you will find exercises for further practice.

You should maintain a Homework Journal. The purpose of the journal is for you to carefully document your work on all homework problems, and to write down questions and comments. The nature of online assignment makes it very important for us to keep a written record of our work. This will help you when it comes time to review for your midterm and final exam. Here are some guidelines for maintaining your homework journal:

- Label each homework problem. Copy down each question and follow it with your solution attempts.
- Your solutions do not need to be perfect. It's okay to make mistakes, cross out work, and start over. In fact, many find it to be useful to review their past mistakes so they can avoid them.
- Write down any definition or theorem you use to in your solution of each problem.

WebAssign:

To enroll in WebAssign, go to <https://www.getenrolled.com/?courseKey=aacc76327741>

This will take you to a Cengage site where the Course Key has already been entered for you. Then click on the ENROLL button and follow the instruction there.

WebAssign problems feature easy links to corresponding material in the online text, as well as links to videos to help students master the material. The interface can recognize many algebraically equivalent expressions. You are provided with five attempts to enter your solution for each free-response problem (one attempt for each multiple-choice problem). Here's a suggestion: if you enter two incorrect answers, it's probably time to seek some help for the problem – with a friend, a tutor, or with me.

Grading Matrix

Your course grade is based on WebAssign homework, written assignments, midterm exam, and the final exam. The course grade, based on your total score out of 100%, will be determined according to the following scale:

90-100 A	80-89 B	70-79 C	60-69 D	0-59 F
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- Turn-around time for graded assignments is 72 business hours from the due time.
- Note that grades can be accessed in Canvas. homework scores will be imported from WebAssign to Canvas at the end of the semester. However, you can find your current grade at any point during the semester using the [Canvas What-If Grades Feature](#).

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Learning Outcomes: By the end of this course, you will be able to:

1. Solve systems of linear equations using a variety of matrix algebra techniques.
2. Use the properties of matrix operations, inverses, and determinants to solve matrix algebra problems.
3. Find the dimension and bases for a variety of vector spaces and subspaces.
4. Use the properties of inner product spaces to create an orthonormal basis, find the angle between vectors, and check for orthogonality.
5. Analyze the properties of transformations in function notation, prove linearity or nonlinearity, and write its matrix representation.
6. Find and apply the eigenvalues and eigenvectors a linear transformation.
7. Prove basic theorems of linear algebra.
8. Apply the techniques of linear algebra to solve problems from the physical and social sciences.

Recording in Classrooms: Unauthorized recording of classroom discussions, presentations or images, including pictures of individuals or instructional materials, or unauthorized reproduction or uploading of such recordings to any web or online environment, is prohibited.

Netiquette Guidelines:

- Always be respectful and show courtesy towards others. Remember that you are communicating with other people. There may be differences of opinion.
- Think before you post and stay on topic. Make sure your commentary or post is relevant to the conversation.
- Use appropriate and respectful language and style with everyone. Avoid the use of ALL CAPS, for example.
- Be considerate of others. Avoid any condescending or derogatory comments. The use of sarcasm is discouraged, as it can be easily misinterpreted in written form.
- Allow for misunderstandings. The nature of nonverbal communication means that there is always a possibility for misinterpretation. Always make the most charitable assumptions about other message's meanings.
- When video conferencing: find a quiet space to talk if possible. I encourage everyone to use your camera during our virtual meetings. You can use a virtual background for privacy if it's helpful. Use the chat feature or raise your hand to ask a question if someone else is talking. Keep your microphone muted if you are not speaking. Remain focused during the meeting.

Technical Assistance: For assistance, call 410-777-4357 (410-777-HELP), email at helpdesk@aacc.edu, or visit the AACC Technology Services at: <http://www.aacc.edu/resources/technology/>

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COVID-19 Policy:

While AACC does not require masking or vaccines, as members of a college community, we ask that all faculty, staff, and students take reasonable precautions to prevent transmissible illnesses like COVID-19. We encourage anyone in the college community to stay home and seek treatment if they are ill. Please contact me if you need assistance with make-up work while you are ill. Please refer to the [COVID-19 Policy webpage](#) for the most up to date policy.

Accommodations

Students who need a reasonable accommodation should contact [Disability Support Services](#) at DSS@aacc.edu or 410-777-1411.

Academic Integrity Policy

In support of this aim, Anne Arundel Community College requires all students to exhibit academic integrity in all their academic work. A culture of academic integrity, a unifying principle at AACC and all academic communities, is built upon respect for others' work, commitment to doing one's own work, and intolerance for academic dishonesty in all its forms. Students are responsible for reviewing and complying with the College's Academic Integrity Policy. A student found in violation of this policy shall be subject to appropriate sanctions, up to and including expulsion from the College, in accordance with the Resolving Allegations of Academic Misconduct Procedure and the Academic Misconduct Hearing Procedure. For more information, visit the [Academic Integrity Policy](#) and [AACC Policies webpage for the Academic Integrity Procedures](#).

Student Conduct Policy

Students shall always conduct themselves in a manner that demonstrates mutual respect and courtesy, displays appropriate standards of behavior, and refrains from any actions or in actions that impinge on the rights of others or disrupt the teaching and/or learning process or the operations of the college. A student found in violation of this policy or any other College policy shall be subject to appropriate sanctions, up to and including expulsion from the College, in accordance with the student conduct procedures. The full text of the policy is available on the college's [AACC Policies webpage](#) and in the Student Handbook and College catalog. Additional information for [Code of Conduct webpage](#).

Acceptable Use of Information Technology Resources Policy

This policy governs the acceptable use of the college information resources by anyone. This policy applies to students enrolled in this course at any time they are using college resources. The goal of the usage policy is to encourage an environment of learning in which all students can interact in an open, legal, and ethical manner. The full text of the policy is available on the AACC [Acceptable Use of Technology Resources Policy webpage](#) procedures that implement the policy are available on the AACC [Acceptable Use of Information Technology Resources Procedures](#).

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Religious Accommodations Policy

Anne Arundel Community College prohibits discrimination, harassment, and retaliation on the basis of religion. The purpose of this Religious Accommodations Policy is to describe the rights of Employees and Students to request and receive Religious Accommodations in accordance with the College's policies and relevant laws and regulations regarding Religious Beliefs. For more information, review the [Religious Accommodations Policy webpage](#). Also, review the [Student Religious Accommodations Procedure webpage](#).

Notice of Nondiscrimination updated June 1, 2023

The college prohibits all types of discrimination including, but not limited to, race – including hair texture and hairstyle – color, religion or creed, ancestry or national origin, sex, age, marital status, physical or mental disability, sexual orientation, gender identity, veteran status, citizenship, and genetic information. Complaints of discrimination shall be reported to the college's chief compliance and fair practices officer. For information regarding the college's compliance and complaints concerning discrimination, contact the chief compliance and fair practices officer and Title IX coordinator at complianceofficer@aacc.edu or 410-777-1239, or call Maryland Relay at 711. For additional information, review the [Equal Opportunity and Nondiscrimination Policy](#).

Americans with Disabilities Act Policy

Anne Arundel Community College prohibits all types of discrimination, harassment, sexual misconduct, and retaliation on the basis of race, including hair style and hair texture, color, religion or creed, ancestry or national origin, sex, age, marital status, physical or mental disability, sexual orientation, gender identity, veteran status, citizenship, and genetic information. To file a complaint of discrimination, harassment, sexual misconduct, or retaliation, please contact the chief compliance and fair practices officer/Title IX coordinator at 410-777-1239 or complianceofficer@aacc.edu. Students who need a reasonable accommodation should contact Disability Support Services at DSS@aacc.edu or 410-777-1411. Requests for most accommodations must be made at least seventy-two (72) hours prior to the event or activity; however, requests for sign language interpreters, live captioning, alternate format books, or assistive technology need to be submitted at least thirty (30) days prior to the event or activity.

Emergency College Closure

If an emergency arises that causes the college to be closed, the planned activity or assignment for that day will occur or be due the next day that classes resume on campus. Faculty and students are required to meet deadlines that occurred prior to the emergency but are not required to meet previously established deadlines during the time the College is closed. Students are strongly encouraged to continue with coursework during a closure to the degree possible, as assignment timelines may be compressed upon the reopening of the college. For closure information, students can check www.aacc.edu, and can sign up for text message alerts using the Stay Informed with Campus Alerts web page. If class is cancelled because of an emergency college closure, please check your course in Canvas for possible announcements.

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Class Expectation/Instructional Hour (Statement on Study Time)

A minimum expectation is that for every hour spent “in class,” as defined by your instructor, you should plan to spend at least two hours “out of class” in preparation. Your instructor or the class may require additional time. More important than how MUCH should someone study is HOW should someone study. Studying is a skill, and if students have not developed that skill, they may still struggle regardless of how much time they study. More information about study skills including time management techniques can be found at:

<http://ola2.aacc.edu/vc/timemanagement/TimeManagementWebShop/>

For a complete list of all college policies, visit <http://www.aacc.edu/policies/>

For a complete list of academic regulations, visit <http://catalog.aacc.edu/> and select Academic Regulations from the menu on the left.

College-wide Core Competencies: Anne Arundel Community College is committed to offering experiences that allow students to acquire, develop and demonstrate growth in 10 core competencies. These competencies encompass general education and essential life skills. Of the ten, this course will focus on providing feedback in the following areas:

- Communication: Using listening, speaking, reading, writing and visual communication skills effectively.
- Technology Fluency: Utilizing technology to enhance productivity in one’s academic, professional and personal life.
- Information Literacy: Recognizing when information is needed and locating, evaluating, and using information appropriately.
- Self-Management: Displaying accountability and adaptability as a learner. Scientific Reasoning Applying logic and the scientific method to interpret observable evidence.
- Quantitative Reasoning: Applying mathematical concepts appropriately to analyze and interpret quantitative information.
- Innovative and Critical Thinking: Integrating knowledge to analyze problems using different modes of thinking (critical, creative, and innovative)